

Robin Boundary Traces Have the Same Approximation Scale as the Poisson Semigroup

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Status. Candidate substantial partial solution, likely valid, for human review. It completely settles the direct/inverse approximation component and the literal reciprocal-resolvent question in the source's open program; it does not claim all of the proposed inverse-ergodic or unbounded generalized-Drazin extensions.

1 Source problem

Butzer and Stens write the Robin boundary trace for the exterior unit ball as a spherical-harmonic multiplier family. Unlike the Dirichlet and Neumann families, it is not a semigroup. They ask for a direct and inverse approximation theorem, a replacement for the missing infinitesimal generator, and an analogue of the resolvent interconnection formula when the operator is not a semigroup generator [1, Section 3, PDF p. 8].

Unfortunately, the family of operators, defined by (8) does not possess the semigroup property, apart from the cases $\beta = 0$ (Dirichlet problem), or $\alpha = 0$ (Neumann problem). This means that one cannot apply the general theory developed in [12] in order to deduce results corresponding to those of Theorems 1.1 and 2.1.

A counterpart of Theorem 1.1(i) $\Leftrightarrow$ (v) or Theorem 2.1(i) $\Leftrightarrow$ (v), thus a direct and inverse approximation theorem for the operators (8), may be proved by classical approximation theoretic methods including K -functional methods, where $\mathcal{D}(A)$ has to be replaced by a suitable subspace of $L^2(S)$. Setting

$$U(r)h(\varphi, \theta) := w(\varphi, \theta, r; h)$$

the operators $(U(r))_{r>1}$ mapping the Hilbert space $L^2(S)$ into $\text{span}\{Y_k^m; k \in \mathbb{Z}, |m| \leq k\}$ are clearly commutative, i. e., $U(r)U(s) = U(s)U(r)$. Hence the new approach presented in [16, 17] can be applied. For the Bernstein-type inequality needed for see, e. g., [18, 19]. See also, [20, pp.], [21, pp.]. For the corresponding Jackson-type inequality see [16, 17, 22], [20, pp.], [21, pp.], [6, Chapter 5]. For the basic spherical harmonic theory in question see, e. g., [6, 7, 19, 23, 3].

For the other equivalent assertions of Part a), however, one has, first of all, to find a replacement for the infinitesimal generator A . With methods of spectral theory (see, e. g., [24, 25, 26]), which enables one to investigate the resolvent of A , one may be able to deduce an equivalence of type (i) $\Leftrightarrow$ (ii).

Concerning (iii) and (iv) of Theorems 1.1 or 2.1, the fundamental tool is the the so-called interconnection theorem, a link between the resolvents of A and A^{-1} , where A is the infinitesimal generator of a semigroup. It reads (see [12]),

$$\lambda R(\lambda; A^{-1})f = f - \lambda^{-1}R(\lambda^{-1}; A)f. \quad (9)$$

The question is, whether there exists something similar, when A is not the generator of a semigroup of operators?

The situation becomes even more complicated, when A is not injective. In this case A^{-1} has to be replaced by the generalized Drazin inverse A^{gd} , introduced by Butzer, Westphal and Kolha (see [1, 27, 2]). Here A is a closed not necessarily bounded operator. If A is the generator of a semigroup, then (9) holds for A^{gd} instead of A^{-1} in a slightly modified form; see [10], [11, p. 37].

Alternative approaches to counterparts of (iii) and (iv) of Theorems 1.1 or 2.1 are using Shaw's theory of $\mathbf{A}$ -ergodic nets [28] or via Hille's pseudoresolvents (see [29, p. 215 ff.], [30, p. 521 ff.]).

The results below give all three objects at the level needed for the approximation question. The key point is that the semigroup law is unnecessary: the Robin multipliers can be compared coefficient by coefficient with the Poisson multipliers.

2 Claimed contribution

Let $H = L^2(S^2)$ and write its orthogonal spherical-harmonic decomposition as

$$H = \bigoplus_{k=0}^{\infty} H_k, \quad h = \sum_{k \geq 0} h_k.$$

Define the positive self-adjoint operator N by

$$Nh_k = (k+1)h_k, \quad \mathcal{D}(N) = \left\{ h : \sum_{k \geq 0} (k+1)^2 \|h_k\|_2^2 < \infty \right\},$$

and let $P_t = e^{-tN}$ be the exterior Poisson semigroup. Fix real Robin parameters α, β , not both zero, satisfying the nonresonance condition

$$\alpha - \beta n \neq 0 \quad (n = 1, 2, \dots). \quad (1)$$

The source's boundary-trace family, with $r = e^t$, is

$$V_t h = \sum_{k \geq 0} b_{k+1}(t) h_k, \quad b_n(t) = e^{-nt} \frac{\alpha - \beta n e^{-t}}{\alpha - \beta n}. \quad (2)$$

This formula is initially read on finite harmonic sums; the uniform estimate proved below extends every V_t , $0 < t \leq t_0$, to a bounded operator on H .

For the usual K -functional we use the equivalent homogeneous graph norm

$$K(t, h; H, \mathcal{D}(N)) = \inf_{g \in \mathcal{D}(N)} (\|h - g\|_2 + t \|Ng\|_2).$$

This is equivalent to the conventional full graph norm because $N \geq I$.

Theorem 1 (Robin direct and inverse theorem). *Under (1), there are $t_0 > 0$ and constants $0 < c \leq C < \infty$, depending only on α, β , such that for every $h \in H$ and $0 < t \leq t_0$,*

$$c K(t, h; H, \mathcal{D}(N)) \leq \|(I - V_t)h\|_2 \leq C K(t, h; H, \mathcal{D}(N)). \quad (3)$$

Equivalently, the Robin error, the Poisson error $\|(I - P_t)h\|_2$, and this K -functional are uniformly comparable.

Consequently, for every $0 < s < 1$,

$$\|(I - V_t)h\|_2 = O(t^s) \iff K(t, h; H, \mathcal{D}(N)) = O(t^s).$$

At the saturation order,

$$\begin{aligned} \|(I - V_t)h\|_2 = O(t) &\iff h \in \mathcal{D}(N), \\ \|(I - V_t)h\|_2 = o(t) &\iff h = 0. \end{aligned}$$

Moreover the strong right derivative of V_t at zero on $\mathcal{D}(N)$ is

$$Ch := \lim_{t \downarrow 0} \frac{V_t h - h}{t} = [-N + \beta N(\alpha I - \beta N)^{-1}] h. \quad (4)$$

The second term is bounded, $\mathcal{D}(C) = \mathcal{D}(N)$, $\|Ch\|_2 \asymp \|Nh\|_2$, and C is a bounded perturbation of $-N$. Thus C is an explicit replacement infinitesimal operator and itself generates a strongly continuous semigroup.

The source's separate resolvent question has an algebraic answer that is independent of the Robin setting.

Proposition 1 (Reciprocal resolvents need no semigroup). *Let $A : \mathcal{D}(A) \subset X \rightarrow X$ be a closed injective operator on a complex Banach space and regard A^{-1} as the closed operator with domain $\text{Ran } A$. If $\lambda \neq 0$ and $\lambda^{-1} \in \rho(A)$, then $\lambda \in \rho(A^{-1})$ and*

$$\lambda R(\lambda; A^{-1}) = I - \lambda^{-1} R(\lambda^{-1}; A). \quad (5)$$

No generator or semigroup assumption is used.

If instead A is a bounded generalized-Drazin invertible operator, let P be its spectral idempotent at zero, $Q = I - P$, and A^D its generalized-Drazin inverse. At reciprocal nonzero resolvent points,

$$\lambda R(\lambda; A^D) = P + Q[I - \lambda^{-1} R(\lambda^{-1}; A)]Q. \quad (6)$$

3 Proof intuition

There are two short mechanisms. First, after dividing by the Poisson error, the Robin error multiplier becomes

$$\frac{1 - b_n(t)}{1 - e^{-nt}} = \frac{nq_n(t) - a}{n - a}, \quad q_n(t) = \frac{1 - e^{-(n+1)t}}{1 - e^{-nt}}, \quad a = \frac{\alpha}{\beta}.$$

Concavity of $x \mapsto 1 - e^{-tx}$ gives $1 < q_n(t) < 1 + 1/n$. Large degrees are therefore uniformly harmless. Only finitely many small degrees remain, and their zero-time limits are nonzero exactly because the Robin problem is nonresonant.

Second, the reciprocal-resolvent formula is simply the factorization

$$(\lambda I - A^{-1})A = \lambda(A - \lambda^{-1}I).$$

The semigroup representation of a resolvent is one way to discover the identity, but is not required to prove it.

4 Proof of Theorem 1

If $\beta = 0$, then $b_n(t) = e^{-nt}$ and the Robin and Poisson errors are identical. Assume $\beta \neq 0$ and put $a = \alpha/\beta$. Condition (1) is precisely $a \notin \{1, 2, \dots\}$.

For $t > 0$ define

$$q_n(t) = \frac{1 - e^{-(n+1)t}}{1 - e^{-nt}}.$$

The function $x \mapsto 1 - e^{-tx}$ is increasing and concave on $[0, \infty)$ and vanishes at zero. Hence

$$1 < q_n(t) < \frac{n+1}{n} = 1 + \frac{1}{n}. \quad (7)$$

A direct calculation from (2) gives

$$\frac{1 - b_n(t)}{1 - e^{-nt}} = \frac{nq_n(t) - a}{n - a}. \quad (8)$$

Choose $n_0 > 2|a| + 2$. For $n \geq n_0$, the two quantities $n - a$ and $nq_n(t) - a$ are positive, and (7) bounds their ratio above and away from zero by absolute constants independent of n, t . For each of the finitely many $1 \leq n < n_0$, the right side of (8) extends continuously to $t = 0$ with value

$$\frac{n + 1 - a}{n - a}.$$

Neither numerator nor denominator vanishes because $a \notin \{1, 2, \dots\}$. Shrinking t_0 and taking the minimum and maximum over these finitely many degrees therefore yields constants $m, M > 0$ such that

$$m(1 - e^{-nt}) \leq |1 - b_n(t)| \leq M(1 - e^{-nt}) \quad (n \geq 1, 0 < t \leq t_0). \quad (9)$$

Orthogonality now gives

$$m\|(I - P_t)h\|_2 \leq \|(I - V_t)h\|_2 \leq M\|(I - P_t)h\|_2. \quad (10)$$

It remains to identify the Poisson error. Introduce the quadratic functional

$$K_2(t, h)^2 = \inf_{g \in \mathcal{D}(N)} (\|h - g\|_2^2 + t^2 \|Ng\|_2^2).$$

Minimizing independently on every H_k gives the exact formula

$$K_2(t, h)^2 = \sum_{k \geq 0} \frac{t^2(k+1)^2}{1 + t^2(k+1)^2} \|h_k\|_2^2. \quad (11)$$

The inequalities

$$K_2(t, h) \leq K(t, h; H, \mathcal{D}(N)) \leq \sqrt{2}K_2(t, h)$$

are immediate. Finally, $1 - e^{-x}$ is uniformly comparable on $[0, \infty)$ with $x/\sqrt{1+x^2}$. Comparing the orthogonal sums for $\|(I - P_t)h\|_2^2$ and (11) proves (3).

The $O(t^s)$ statement follows immediately. Also $K(t, h) \leq t\|Nh\|_2$ for $h \in \mathcal{D}(N)$. Conversely, if $K(t, h) = O(t)$, divide (11) by t^2 and use Fatou's lemma as $t \downarrow 0$ to obtain $\sum (k+1)^2 \|h_k\|_2^2 < \infty$. If $K(t, h) = o(t)$, the same formula applied to each fixed k forces every $h_k = 0$. This proves the saturation assertions.

For the derivative, differentiating (2) at zero gives the multiplier

$$c_n = -n + \frac{\beta n}{\alpha - \beta n}.$$

The rational second term is bounded under (1), while $|c_n| \asymp n$: after division by n its absolute value is $|(n+1-a)/(n-a)|$, a nonzero sequence converging to one. The scalar bound (9) also gives $|(b_n(t) - 1)/t| \leq Mn$. Dominated convergence in the orthogonal sum now proves (4) for $h \in \mathcal{D}(N)$. The domain and norm assertions follow, and the bounded perturbation theorem gives the final generator statement.

5 Proof of Proposition 1

Put $\mu = \lambda^{-1}$. On $\mathcal{D}(A)$,

$$(\lambda I - A^{-1})A = \lambda(A - \mu I).$$

Since $\mu \in \rho(A)$, this factorization shows that $\lambda I - A^{-1} : \text{Ran } A \rightarrow X$ is bijective. Solving the factorization and using $(A - \mu I)^{-1} = -R(\mu; A)$ gives

$$R(\lambda; A^{-1}) = \lambda^{-1}I - \lambda^{-2}R(\lambda^{-1}; A),$$

which is bounded and is equivalent to (5).

For the generalized-Drazin statement, use the reducing decomposition $X = PX \oplus QX$. On PX , $A^D = 0$, so the left side of (6) equals P . On QX , A is invertible and $A^D = A^{-1}$, so (5) applies. Combining the two blocks proves (6).

6 Verification, scope, and novelty

The separate report `verification.md` audits the resonance condition, uniformity in the harmonic degree, the K -functional converse, and the closed-operator domains. The script in `code/` checks the scalar factorization, derivative, resolvent identity, and representative numerical ratios; computation is not used as proof.

The theorem gives the requested direct/inverse approximation counterpart for all real nonresonant Robin parameters, not merely the opposite-sign example mentioned in the source. Proposition 1 answers the literal injective resolvent question and gives the standard bounded generalized-Drazin modification. We do not claim counterparts of every inverse-ergodic assertion in the source, nor an unbounded a -Drazin theorem.

A bounded search on 11 August 2026 covered the run's four cheap indexes, exact-title and arXiv searches for arXiv:1901.04450, the displayed resolvent formula, the source authors' later exterior-boundary work, and combinations of "Robin", "K-functional", and "spherical harmonic multiplier". A 2022 follow-up found in the search treats exterior Dirichlet and Neumann problems, not this Robin comparison. No later source stating the theorem above was found. This supports, but cannot certify, novelty. The resolvent part itself should be regarded as an elementary clarification, not as a new operator identity.

References

- [1] P. L. Butzer and R. L. Stens, *The Third Boundary Value Problem of Potential Theory for the Exterior Ball and the Approximation Behaviour of the Solution; a Novel Open Problem*, arXiv:1901.04450 (2019), <https://arxiv.org/abs/1901.04450>.